

# Aryan Gahlot

Nagpur, Maharashtra, India | aryangahlot50@gmail.com | linkedin.com/in/aryansg2 | Portfolio

## About

building software at the intersection of AI and backend engineering. Interested in LLMs, intelligent agents, APIs, distributed systems, and developer tools

## Education

**Shri Ramdeobaba College of Engineering and Management**

*B.E., Electronics and Communication Engineering*

*Coursework: Signal Processing, Linear Algebra, Probability & Statistics, Data Structures, DBMS*

Expected May 2027

*Nagpur, India*

## Technical Skills

**Frameworks & NLP:** PyTorch, TensorFlow, Hugging Face, Keras, Scikit-learn, OpenCV, Pandas, NumPy

**Languages:** Python, C++, MATLAB, SQL, R, Bash,  $\text{\LaTeX}$

**Software Development:** Java (Spring Boot, Spring Cloud), .NET 8, REST APIs, Microservices, TypeScript/Bun, Vue 3, WebSocket/SSE

**Backend & Infrastructure:** Docker, Kubernetes, Kafka, RabbitMQ, Nginx, AWS (EKS), GCP (GKE), Azure (AKS), Terraform, Helm, CI/CD

**Tools:** MLflow, Git, Jupyter, Google Colab, PostgreSQL, Snowflake, MinIO

**ML Concepts:** Transformers, LLMs, NLP, Attention Mechanisms, Model Compression, Evolutionary Algorithms, Contrastive Learning, Multi-Agent Systems, Uncertainty Quantification, Time-Series

**Mathematics:** Linear Algebra, Optimization Theory, Probability & Statistics, Algorithm Analysis

## Projects

**FluxPipe** – *Continuous Data Flow Platform*

2025

- Built a production-grade, fully containerized data platform orchestrating 20 Docker services – spanning batch ingestion (MySQL → Airflow → Spark → Snowflake/PostgreSQL), real-time Kafka streaming with anomaly detection, MLflow experiment tracking, and end-to-end observability via Prometheus and Grafana.
- Engineered a star-schema data warehouse with Snowflake as the primary store and PostgreSQL as a local development fallback; implemented hourly ETL DAGs for dimension and fact loading alongside daily batch and streaming monitoring pipelines orchestrated through Apache Airflow.
- Deployed across AWS EKS (Terraform + Helm), GCP GKE, Azure AKS, and on-premises Kubernetes clusters via a single Makefile target; developed a .NET 8 REST API exposing 20 endpoints with Polly retry policies, Serilog structured logging, and interactive Swagger documentation.
- Validated end-to-end pipeline integrity with a 35-test suite covering Kafka producer logic, Great Expectations data-quality rules, warehouse DDL, Docker Compose service definitions, and Snowflake schema integrity.

**Redag** – *AI Workflow Orchestration Platform*

2025

- Designed an enterprise-grade AI workflow automation platform enabling users to compose reusable DAG pipelines through a drag-and-drop Vue Flow canvas, chaining file uploads, OCR, transcription, summarization, translation, and model inference into auditable, production-ready pipelines.
- Architected a Spring Cloud microservices backend (Java 17 / Spring Boot 3.2) across seven services – auth, workflow, task, AI, file, notify, and gateway – integrated via Nacos service discovery, Sentinel rate limiting, Seata distributed transactions, RabbitMQ async task queues, and MinIO object storage.
- Implemented a Python AI service handling media processing and model inference (faster-whisper, FFmpeg, OpenAI SDK, Ollama), with real-time pipeline status streamed to the Vue 3 frontend over WebSocket and SSE; Docker Compose + Nginx provides a one-command deployment path.

**Heisenberg** – *Agentic Financial Research Terminal*

2025

- Built a production-grade autonomous NLP query routing system (TypeScript/Bun, Docker, AWS) classifying natural language financial queries through 8 hedge-fund strategy lenses with a real-time alert engine and conviction scoring; demonstrates applied agentic systems deployment.
- Implemented sub-100 ms end-to-end query resolution with multi-source financial data aggregation and structured strategy-lens output formatting.

## Work Experience

**IIM Nagpur**

*Research Intern*

Jan 2025 – Present

*Nagpur, India*

- Investigating deep reinforcement learning and neural combinatorial optimization approaches to the Vehicle Routing Problem (VRP) in logistics; framing capacitated VRP variants as sequential decision-making tasks solved via attention-based encoder-decoder architectures.
- Benchmarking Transformer-based solvers and Pointer Network models against classical OR heuristics (Clarke-Wright, LKH3) on real-world supply-chain datasets to quantify solution quality and computational trade-offs.